

# Lachlan Watts-Tobin

(707) 601-5830 • lachlanwt@berkeley.edu • linkedin.com/in/lachlanwt • lachlanwt.com

## EDUCATION

### University of California, Berkeley

Berkeley, CA

B.S. Electrical Engineering & Computer Science, Minor in Mechanical Engineering

May 2028

**Coursework:** Digital Design and Integrated Circuits (with ASIC Lab), Computer Architecture, Internet of Things, Machine Structures, Circuits and Devices, Data Structures, Signals and Information Processing, Solid Mechanics

## SKILLS

**Languages:** SystemVerilog, Verilog, RISC-V, Chisel, C, C++, Rust, Java, Python, MATLAB, SQL, HTML, CSS

**Platforms:** ASIC, FPGA, Git, KiCad, Altium, LTspice, SolidWorks, Fusion 360, OpenCV, PyTorch, Pandas, NumPy

## EXPERIENCE

### Pano AI

San Francisco, CA

Hardware Engineering Intern

May 2026 – August 2026

- Redesigned an edge-computer system layout to fit all electrical components within tight space constraints.
- Authored standard operating procedures and designed a manufacturing plan to scale edge-computer production.
- Developed real-time lens-contamination detection in Python/OpenCV using FFT and Laplacian variance analysis.

### Vision IV

(Remote) Yokohama, Japan

Hardware Engineering Intern

May 2025 – July 2025

- Optimized a high-performance electrical component using experimental materials to improve reliability.
- Analyzed potential failure points to guide data-driven improvements to component performance.

### UC Berkeley, College of Engineering

Berkeley, CA

Mathematics Instructor

June 2025 – August 2025

- Taught pre-calculus to ~100 incoming engineering students in a collaborative learning environment.
- Developed a rigorous pre-calculus curriculum with original lessons and assignments.

## PROJECTS

### CS61CPU

UC Berkeley, College of Engineering, (November 2025 – December 2025)

- Designed a 5-stage pipelined CPU in Logisim capable of executing 32-bit RISC-V instructions.

### Watts & Cseh Electronics

Co-Founder & Lead Electrical Engineer (May 2026 – Present)

- Created and verified analog circuits for guitar effects pedals, inspired by vintage hardwired models.

### AI-Powered Guitar Pedal

Hackathons at Berkeley, (June 20, 2026)

- Bridged AI to hardware, pushing generated parameters from plain English to an ESP32 DSP device over WiFi.

### Automated Trading Card Sorter

UC Berkeley, College of Engineering, (April 2026 – May 2026)

- Engineered a machine that automatically sorts trading cards by their live market value, using CV and scraping.

### Self-Hosted Portfolio Website

Personal, (May 2026 – August 2026)

- Self-hosted a portfolio site on a Raspberry Pi via a Cloudflare with custom DNS and automated Git deploys.

### Filament Winder G-code Slicer

UC Berkeley, College of Engineering, (January 2025 – April 2025)

- Developed standalone software to generate G-code from part files, visualize it, then send it to a custom winder.

## EXTRACURRICULARS

### Bike Builders of Berkeley

Berkeley, CA

Lead Electrical Engineer

August 2026 – Present

- Lead the electrical design of the e-bike's EV drivetrain, including custom PCBs and power electronics.
- Balance workloads across website, firmware, and hardware as project manager.

### Engineering Student Services

Berkeley, CA

Undergraduate Tutor

August 2026 – Present

- Tutor lower-division electrical engineering and computer sciences coursework to students every weekday.